Predictive Rate and Structural-Use Opacity in Explanatory-Primitiveness Judgments: A Staged Bidirectional Neural Test

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Abstract

Why do some perceptual qualities seem not merely difficult to describe but resistant to decomposition into relations, transformations, and causes? This manuscript proposes a restricted **Predictive Rate–Structural Opacity Hypothesis** about reports of explanatory primitiveness. It does not propose a measure of phenomenal occurrence. The hypothesis states that such reports become more likely when a first-order neural representation retains relatively little information about a task-defined sensorimotor history while preserving predictive utility, and when relational information represented in that first-order state cannot be used by the participant in novel structural and counterfactual tasks.

The proposal is motivated by Automated Compression Theory, which argues that compressed first-person access may contribute to the appearance of exceptional phenomenal properties [1], by predictive-processing accounts of hierarchical generative prediction [2], and by information-dynamic approaches emphasizing prediction and efficient re-representation [4]. The revision separates representational rate from predictive utility. Let X^+ be a finite, preregistered sensorimotor-history representation, Y a finite future target, and Z an operational first-order neural representation. Representational rate and utility are

$$r_Z = I(Z; X^+) \qquad \text{and} \qquad u_Z = I(Z; Y).$$

Compression amount is

$$K_Z = 1 - \frac{r_Z}{H(X^+)},$$

interpreted only when predictive utility, structural adequacy, reliability, and distance-to-frontier gates are satisfied. The primary manipulation check compares r_Z between regularity conditions at matched u_Z; it does not use the former ratio $I(Z;Y)/I(Z;X)$.

Structural opacity is behaviorally anchored. A finite query Q specifies a relational or counterfactual problem, and $T = \tau(G, Q)$ is its finite correct answer. First-order structural adequacy is

$$a_Z = \frac{I(T; Z \mid Q)}{H(T \mid Q)}.$$

Participant-level executable structural competence e_{use} is estimated from accuracy and proper scoring on unannounced, nonverbal, novel-state transfer tasks. The primary opacity score is

$$\Omega_{\text{use}} = a_Z\,(1 - e_{\text{use}})\,.$$

It is high only when the first-order representation carries task-relevant relational structure and the participant cannot use that structure. Conditional external decoding from a metacognitive neural state M remains a secondary convergent diagnostic, normalized by fixed $H(T \mid Q)$, not a sufficient measure of internal executability.

A staged program first validates measurement reliability, task internalization, online paired Z and M estimation, and TMS target engagement. The confirmatory experiment then crosses synthetic-world regularity with structurally useful training while using multiple independent world exemplars per cell. A second experiment applies participant-specific, sham-controlled stimulation during concurrent high-density EEG. The primary report outcome is structural-explanation primitiveness; ineffability and intrinsicness are modeled as potentially distinct outcomes. The principal prediction is a positive $K_Z \times \Omega_{\text{use}}$ interaction, with predictive utility modeled separately. Instruction should improve structural use and lower primitiveness particularly for compact, predictively adequate codes. Temporally targeted disruption should reverse those changes in targetable participants while leaving first-order performance, semantic retrieval, and the measured Z representation within preregistered equivalence bounds.

Confirmation would support a causal account of reported structural primitiveness under specified metacognitive conditions. It would not show that compression constitutes consciousness, that mastering external-world rules explains why experience exists, that posterior neural geometry fixes phenomenology, or that a metaphysical hard problem has been resolved.

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1. Revised Research Question and Scope

1.1 Primary question

The primary question is:

> **When task-level sensory histories, predictive utility, and first-order performance are matched, are reports that a perceptual quality is structurally primitive associated with the conjunction of low first-order representational rate and poor participant-level access to relational information carried by that representation?**

This wording is deliberately narrower than asking whether consciousness itself arises from compression or whether the metaphysical hard problem has a neural solution. The study directly concerns reports made by trained adult participants about synthetic visual–haptic perceptual categories. It manipulates one form of metacognitive access, so it does not claim to hold “cognitive access” in general fixed. The intended invariances concern external test sequences, first-order task performance, measured predictive utility, global reportability, and selected properties of the first-order neural representation.

The central outcome is **reported structural-explanation primitiveness**: endorsement that a perceptual quality remains an unanalyzable or explanatory stopping point even when its similarities, contrasts, transformations, and action consequences are considered. That outcome is linguistically mediated. It is not an appearance observed independently of report, and matching global reportability does not make it report-independent.

1.2 Three claims that must remain separate

The manuscript distinguishes three propositions:

1. **Phenomenal occurrence:** a state is experienced.
2. **Phenomenal organization:** an experienced state has similarity, contrast, temporal, bodily, affective, and perspectival structure.
3. **Explanatory-primitiveness report:** a participant reports that the state’s qualitative character resists structural reconstruction or explanation.

Only the third proposition is directly tested. Selected behavioral tasks probe aspects of the second, but they do not establish the complete structure of phenomenology. The first is presupposed in the ordinary experimental sense that participants report visual–haptic experiences; it is not measured by K_Z, Ω_{use}, or any neural decoder.

A change in explanatory-primitiveness reports would therefore not show that experience disappeared, became less vivid, or ceased to possess whatever metaphysical status it had. Conversely, an unchanged report would not establish ontological primitiveness.

1.3 The restricted architectural hypothesis

The revised hypothesis contains four components:

1. A first-order representation Z can retain a relatively low rate of information about a defined sensorimotor history X^+.
2. The same representation can preserve nontrivial predictive utility about a future target Y.
3. Z can represent a finite relational or counterfactual target T that the participant nevertheless cannot use effectively.
4. The conjunction of compact predictive coding and failed structural use increases reported explanatory primitiveness.

The distinctive statistical claim is

$$\beta_{K\Omega} > 0$$

in a model that includes compression amount, structural-use opacity, predictive utility, frontier distance, assignment effects, and hierarchical variation.

The proposal does not require that compression alone increase primitiveness. A compact representation could be accompanied by excellent structural access. It does not require that poor structural use alone increase primitiveness. Failure on a novel task may be experienced as ordinary uncertainty or ignorance. The predicted report profile is strongest when the first-order state is compact, useful, structurally adequate, and unavailable for participant-level relational use.

1.4 A separate hard-problem bridge

A broader philosophical bridge is treated as a distinct, secondary hypothesis:

> If the same manipulation that alters trained-world structural-primitiveness reports also changes delayed responses to independently developed hard-problem-like vignettes, without teaching the language used in those vignettes, then the local mechanism may contribute to a wider class of phenomenal explanatory judgments.

This bridge can fail even if the primary experimental hypothesis succeeds. A local effect on trained perceptual judgments would not automatically generalize to conceivability arguments, knowledge arguments, ontological commitments, or the claim that physical processing is accompanied by experience. The manuscript retains possible relevance to those debates while rejecting the inference that the experiment directly tests metaphysical truth.

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2. Theoretical Motivation and Evidential Boundaries

2.1 Automated compression and exceptional-property judgments

Automated Compression Theory proposes that first-person access operates over compressed sensory information and that the distorting character of such access can make consciousness appear to possess exceptional properties [1]. Its explanatory strategy is etiological: explain why people form hard-problem intuitions rather than infer metaphysical exceptionalism directly from how experience appears.

The present proposal adopts that etiological orientation but revises the operational bridge. Compression cannot be inferred from a ratio expressing how much retained information is predictive. A compact sufficient code and an uncompressed lossless code can both retain only predictive information. Representational rate must therefore be estimated separately from predictive utility.

The present account also adds a structural-use condition. A compact code need not be opaque. A short equation can be both efficient and transparent to someone who knows how its variables generate outcomes. Conversely, a representation can be inaccessible because it is noisy, forgotten, or poorly learned without being compressed. Reports of intrinsic or unanalyzable qualitative identity are therefore predicted to depend on a conjunction rather than compression alone.

2.2 Predictive processing without an assumed neural bottleneck

Predictive-processing accounts describe perception and action in terms of hierarchical generative models, top-down prediction, bottom-up prediction error, and recurrent cortical exchange [2]. This framework motivates choosing a future-relevant variable Y rather than treating compression as indiscriminate loss of sensory detail.

A forward-looking theory of content likewise emphasizes an agent's capacity to detect and correct prediction errors, thereby preserving the possibility of misrepresentation while grounding determinate content in future-directed capacities [21]. In the proposed task, the utility of Z is evaluated against outcomes and transformations that matter for action in the synthetic world.

These theoretical sources do not establish that a biological neural representation minimizes a particular information-bottleneck objective. Machine-learning work formalizes a trade-off between information retained about an input and information useful for predicting a target [24]. Other learning-theoretic work argues that controlling information bottlenecks can support generalization in specified deep-learning settings while stressing that such control is neither necessary nor the only route [23]. Those results motivate a rate–utility analysis but do not validate biological implementation, neural mutual-information estimation, or a theory of consciousness.

2.3 Re-representation and format mismatch

Information Dynamics of Thinking describes the mind as predicting its world and repeatedly re-representing information to improve efficiency [4]. Re-representation allows first-order and metacognitive states to differ in format. A first-order representation may support rapid discrimination and action without making the relations responsible for those capacities available to deliberate counterfactual use.

The revised hypothesis does not require the metacognitive state M to contain a copy of Z. Nor does it require M to contain the currently decoded value of every latent variable. It requires evidence that the participant can mobilize a transformation from the perceptual state and task query to a correct novel structural response. Participant behavior, temporal availability, and neural interaction are therefore more central than an external decoder's success.

2.4 A distributed computational boundary

A computational-boundary account characterizes a self by the spatiotemporal events it can measure, model, and attempt to affect, without requiring a single anatomical self module [3]. The present use of M follows this functional approach. M is not a homunculus or internal spectator. It is an operationally identified set of distributed processes involved in structural use, confidence regulation, self-attribution, semantic access, and report preparation.

Opacity is consequently system- and task-relative. Information may be present in a neural representation but unavailable for the participant's current structural action. Training can expand the relevant computational boundary; fatigue, interference, or perturbation can contract it. None of these changes determines whether the represented state is experienced.

2.5 Embodied qualification

An embodied approach to mind emphasizes body schema, proprioception, intermodal perception, ownership, agency, and the structures of experience that may operate before explicit awareness [12]. The present task is visual–haptic rather than exclusively visual for this reason. Synthetic objects resist action, produce force feedback, and support cross-modal prediction.

Even so, the task captures only a narrow part of embodiment. It does not model pain, homeostatic urgency, affective appraisal, bodily ownership, social understanding, or temporal selfhood. The information-theoretic variables introduced below are not proposed as exhaustive descriptions of embodied experience.

2.6 Comparative and evolutionary restraint

One evolutionary proposal assigns phenomenal properties a labeling role in discriminating and evaluating mental representations and suggests that some animals may share phenomenal consciousness [10]. The present framework is compatible with the functional usefulness of compact perceptual labels, but its report outcome is specific to linguistically competent humans.

Implementation-agnostic work on cognition in diverse embodiments cautions against treating familiar human performance as the sole form of intelligence or agency [27]. Related work emphasizes teleonomic competence in chimeric and synthetic organisms across differences of composition and origin [28]. These perspectives support caution, not an inference that the present metrics detect consciousness in animals or artificial systems.

2.7 Citation scope

The stored sources support broad philosophical and computational motivations. They do not support the exact definitions of K_Z, a_Z, e_{use}, or Ω_{use}; those are novel proposals. They also do not validate source-resolved EEG, trial-level neural information estimation, fMRI-constrained TMS targeting, concurrent TMS–EEG artifact control, psychometric measurement of explanatory primitiveness, or the proposed numerical thresholds. The manuscript therefore treats all such components as methods requiring independent validation rather than as established techniques licensed by the cited theoretical literature.

Metadata-only chapter records are not used as substantive support. Bibliographically anomalous or secondary records are likewise omitted where stronger stored support is unavailable.

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3. Explananda and Confirmatory Hypotheses

3.1 Primary report outcome

The primary report construct is **structural-explanation primitiveness**, denoted P_{str}. It concerns whether the participant treats a current perceptual category as an unanalyzable qualitative unit after considering transformations, similarities, and action consequences.

Candidate items include:

- “This quality seems to remain an unanalyzable unit.”
- “Two states could match in all the learned relations and still differ in the respect that matters here.”
- “The relations among this state and the alternatives seem insufficient to reconstruct its particular character.”
- “If all of the state’s transformations and action consequences were known, nothing structurally important would remain unspecified.” This item is reverse scored.

The wording used in instruction will not contain these sentences or their distinctive philosophical terms. Items will be developed by a team blinded to condition-specific training materials and cognitively interviewed in a separate sample.

3.2 Distinct secondary outcomes

Two related but nonidentical outcomes are retained:

- P_{ineff}: reported difficulty of adequate description or reconstruction;
- P_{intr}: reported independence of the quality from relations, alternatives, actions, and causes.

These outcomes may dissociate. Ineffability can result from limited vocabulary; intrinsicness can be endorsed despite fluent description; structural-explanation resistance can persist even when relational structure is understood. The confirmatory psychometric model will therefore begin with three correlated factors rather than presuming a single reflective factor.

A general factor P will be reported only if:

1. the correlated-factor and bifactor models support a stable common dimension;
2. item thresholds and loadings are sufficiently invariant across assignments;
3. reverse-worded items do not form a dominant method factor; and
4. the common score predicts independently collected transfer judgments.

Failure of unidimensionality is theoretically informative, not automatically a measurement failure. The primary hypothesis can be confirmed for P_{str} while failing for ineffability or intrinsicness.

3.3 Behavioral structural use is not the report outcome

Participant-level structural use is measured through unannounced tasks rather than self-assessment. Examples include:

- selecting which novel state would result from a nonverbal transformation diagram;
- reconstructing a missing state in a relational array;
- choosing a haptic consequence for a previously unseen visual combination;
- identifying which intervention would reverse a state transition;
- generalizing to new symbols and combinations not used in training.

These tasks produce the executable-use score e_{use}. They are not included in P_{str}. This separation prevents the opacity measure from being a paraphrase of the dependent report.

3.4 Main hypotheses

The confirmatory hypotheses are:

- **H1: Rate manipulation.** High generative regularity lowers representational rate r_Z relative to low regularity at matched predictive utility u_Z, with first-order performance modeled separately.
- **H2: Structural-use acquisition.** True structural training increases e_{use}, increases metacognitive target information a_M, and lowers Ω_{use}, while leaving a_Z and the preregistered first-order Z geometry within equivalence bounds.
- **H3: Continuous interaction.** P_{str} increases with the interaction $K_Z\Omega_{\text{use}}$ after predictive utility, rate–utility frontier distance, world family, and assignment are modeled.
- **H4: Randomized assignment interaction.** The effect of structural training on P_{str} is more negative in high-regularity than low-regularity worlds.
- **H5: Reverse perturbation.** In participants with prospectively validated targets, active metacognitive-site stimulation reduces online structural use and a_M, raises Ω_{use}, and shifts P_{str} toward the pretraining range more strongly in compact-code worlds than sham or control-site stimulation.

- **H6: Selectivity.** The effects in H2 and H5 occur without a meaningful reduction in first-order accuracy, measured predictive utility, semantic retrieval, global reportability, or the defined Z representation.
- **H7: Philosophical bridge.** Training-related change in P_{str} predicts delayed change on independently developed hard-problem-like vignettes. This is a secondary hypothesis and is not required for confirmation of H1–H6.

4. Formal Framework

4.1 Variables and mathematical type

All primary task variables are finite and discrete at preregistered resolution.

Symbol	Definition
s	Participant.
w	Synthetic-world exemplar.
t	Trial.
R	Randomized regularity assignment: high or low factorization.
D	Randomized training assignment: true structural training or matched fluent control.
H	Stimulation assignment: active metacognitive target, sham, or active control site.
G	Finite latent state of the synthetic world.
X^{+}	Finite task-defined sensorimotor and context history available before prediction.
Y	Finite future action consequence or transition target.
Q	Finite structural query.
$T = \tau(G, Q)$	Finite correct answer to query Q.
Z	Operational first-order neural representation.
M	Operational metacognitive neural representation.
A	Participant response to a structural query.
r_Z	Representational rate $I(Z; X^{+})$.
u_Z	Predictive utility $I(Z; Y)$.
K_Z	Compression amount relative to $H(X^{+})$.
d_F	Distance from a preregistered rate–utility frontier.
a_Z	First-order structural adequacy.

Symbol	Definition
a_M	Metacognitive target information.
e_{use}	Participant-level executable structural competence.
Ω_{use}	Behaviorally anchored structural-use opacity.
P_{str}	Structural-explanation-primitiveness report score.
P_{ineff}	Ineffability report score.
P_{intr}	Intrinsicness report score.
Γ_Z	Preregistered representational geometry of Z.
Q_{glob}	Global reportability.
S	Sensory surprise.
K_{conf}	Confidence.
V	Vividness.

The finite coding of G, X^+, Y, Q, and T avoids interpreting differential entropy as a bounded fraction. A physically continuous compliance variable may exist in the apparatus, but the confirmatory target is its preregistered finite category. Continuous analyses are secondary and use proper scores or rate–distortion quantities rather than division by differential entropy.

4.2 Complete history representation

The original simple graph $Z \leftarrow X \rightarrow Y$ is inadequate for recurrent perception. The revised X^+ contains:

- the current sensory sequence at fixed resolution;
- recent actions and haptic outcomes;
- current world cue;
- trial position;
- learned transition history;
- a preregistered behavioral estimate of the participant's current prior or belief state;
- relevant context available before Y.

The representation remains incomplete because internal memory and priors cannot be exhaustively observed. The primary compression statistic below does not require

$$Z \perp\!\!\!\perp Y \mid X^+.$$

Instead, the possible violation

$$\lambda = I(Z; Y \mid X^+)$$

is estimated as a diagnostic. A large λ indicates that measured X^+ omits future-relevant influences on Z. Such a result limits interpretation of the rate–utility analysis but does not make the compression amount algebraically undefined.

4.3 Representational rate and predictive utility

Representational rate is

$$r_Z = I(Z; X^+).$$

It measures how much information the operational neural representation retains about the task-defined history. Predictive utility is

$$u_Z = I(Z; Y).$$

The two quantities are reported separately. A representation can have:

- low rate and low utility, indicating possible impoverishment;
- high rate and high utility, indicating informative but noncompact coding;
- low rate and high utility, indicating compact predictive coding;
- high rate and low utility, indicating retention of task-irrelevant detail.

No ratio of u_Z to r_Z is used as a compression measure.

4.4 Compression amount

Because X^+ is finite,

$$0 \le I(Z; X^+) \le H(X^+).$$

When

$$H(X^+) \ge \delta_X > 0,$$

compression amount is defined as

$$K_Z = 1 - \frac{I(Z; X^+)}{H(X^+)} = 1 - \frac{r_Z}{H(X^+)}.$$

At the population level,

$$0 \le K_Z \le 1.$$

A lossless copy of X^+ has $K_Z = 0$, even if every retained bit predicts Y. A one-bit code can have high K_Z, but it does not qualify as an adequate compact predictive representation unless it also passes utility and structural-adequacy gates.

The primary regularity test does not depend solely on this normalized score. It compares r_Z across conditions at matched u_Z, matched entropy of X^+, and matched first-order performance. This rate-at-utility contrast is the main operational test of compression.

4.5 Rate–utility frontier

For each world family, define a preregistered feasible encoder class $\mathcal{E}$. Its task-specific rate–utility frontier is

$$r^*(u) = \inf_{\tilde{Z} \in \mathcal{E}} \left\{ I(\tilde{Z}; X^+) : I(\tilde{Z}; Y) \geq u,\ a_{\tilde{Z}} \geq a_{\min} \right\}.$$

Here $a_{\tilde{Z}}$ is the structural adequacy of a candidate representation, and $a_{\min}$ is fixed before confirmatory analysis. The normalized excess-rate statistic is

$$d_F = \frac{r_Z - r^*(u_Z)}{H(X^+)}.$$

A representation near the frontier uses approximately the minimum modeled rate needed for its predictive and structural utility. A high K_Z accompanied by low utility or large frontier distance is not interpreted as efficient compression.

The frontier is model-relative. It is not the unknowable optimum over every biologically possible code. It will be estimated from the known world generator, preregistered stochastic encoders, and held-out neural mappings. Raw estimates of d_F may fall below zero because of estimation error; they will not be clipped.

4.6 Relational target and internalization gate

The synthetic world has external causes, but the intended construct is not mere knowledge of external rules. For each query Q, the target

$$T = \tau(G, Q)$$

specifies a finite relation or counterfactual concerning the perceptual state. Examples are:

- which state is nearest under the participant's learned similarity organization;
- which transformation changes one perceptual category into another;
- which haptic consequence follows after a novel intervention;
- which missing state completes a relational pattern;
- which transformation reverses the currently represented transition.

A target is treated as internalized only if it passes the following gate:

1. T is decodable from Z on held-out trials;
2. T-based relational distances predict Γ_Z beyond raw stimulus features;
3. the effect generalizes across superficial symbol remappings;
4. decoding remains after adjustment for motor choice, word identity, confidence, and world identity;
5. the result replicates across independent world families.

This gate establishes that the experimentally defined structure is represented in the operational first-order neural state. It does not establish that T exhausts the neural state's organization or causes its phenomenal character.

4.7 First-order structural adequacy

Structural adequacy is

$$a_Z = \frac{I(T;Z \mid Q)}{H(T \mid Q)}.$$

Because T is finite and

$$H(T \mid Q) \geq \delta_T > 0,$$

the population quantity satisfies

$$0 \leq a_Z \leq 1.$$

A high value means that Z discriminates the correct structural answer conditional on the query. It does not imply that the participant can use that information.

4.8 Executable structural competence

Participants respond to novel structural queries with a categorical choice and a probability distribution over possible answers. Let ℓ_{human} be the expected proper log loss assigned to the true target, $\ell_0 = H(T \mid Q)$ the loss of the fixed query-conditional baseline, and ℓ_{oracle} the preregistered finite-precision oracle loss. Executable competence is

$$e_{\text{use}} = 1 - \frac{\ell_{\text{human}} - \ell_{\text{oracle}}}{\ell_0 - \ell_{\text{oracle}}}.$$

Under better-than-baseline, no-better-than-oracle population performance,

$$0 \leq e_{\text{use}} \leq 1.$$

Finite-sample estimates may be negative or exceed one. They will be reported without truncation. A preregistered validity gate requires that the population estimate not be materially below baseline before the bounded construct interpretation is used.

The primary estimate combines several task classes in a hierarchical model. Accuracy, response time, and probability calibration are reported separately so that a single scoring convention cannot determine the result. Performance on familiar trained examples is excluded from the primary estimate; only unannounced, novel-state and counterfactual trials count.

4.9 Behaviorally anchored structural-use opacity

The primary opacity score is

$$\Omega_{\text{use}} = a_Z \left(1 - e_{\text{use}}\right).$$

This score has the desired monotonic properties within its validity region:

$$\frac{\partial \Omega_{\text{use}}}{\partial a_Z} = 1 - e_{\text{use}} \geq 0$$

and

$$\frac{\partial \Omega_{\text{use}}}{\partial e_{\text{use}}} = -a_Z \leq 0.$$

Thus opacity increases when more relational information is represented in Z while participant-level use remains poor, and it decreases as executable use improves. If $e_{\text{use}} = 0.99$ and $a_Z = 1$, opacity is 0.01, not maximal. If participant use becomes sufficient, $\Omega_{\text{use}} = 0$ rather than becoming undefined.

The score is a construct index, not a theorem about unique information. Multiplication assumes that represented structure and use competence are commensurable after normalization. That assumption is tested through convergent behavioral and neural measures rather than treated as self-evident.

4.10 Metacognitive neural information

The metacognitive neural representation is assessed by

$$a_M = \frac{I(T; M \mid Q)}{H(T \mid Q)}.$$

A secondary neural gap is

$$g_M = a_Z - a_M.$$

This difference can be negative and will not be clipped. It compares amounts of target information, not the identity of represented bits. A high a_M does not prove that M implements a decoder, because M may encode a label without supporting structural use. A low a_M does not prove absence of an internal decoder, because M might carry control parameters or use a transformation outside the measurement model.

The neural measure is therefore convergent evidence only. The central opacity construct remains anchored in what the participant can do with novel structure.

4.11 Common prequential code for external decoding

External conditional decoding is retained as a diagnostic rather than the definition of opacity. All targets are finite. The M-only and M, Z models use a common blockwise prequential code.

Let $B_1, \ldots, B_J$ be temporally ordered blocks fixed before fitting. The first block is encoded with the query-conditional baseline. For later blocks, a model is trained only on previous blocks. The total code for T given M is

$$L_M = L(\mathcal{A}_M) + L(\mathcal{H}_M) + \sum_{i \in B_1} -\log p_0(t_i \mid q_i) + \sum_{j=2}^{J} \sum_{i \in B_j} -\log q_{\hat{\theta}_{M,<j}}(t_i \mid m_i, q_i),$$

where:

- $L(\mathcal{A}_M)$ is the code length for the model-family index;
- $L(\mathcal{H}_M)$ is the code for the finite hyperparameter choice;
- p_0 is the fixed baseline code;

- $\hat{\theta}_{M,<j}$ is fitted only on blocks preceding B_j.

The M, Z code is

$$L_{MZ} = L(\mathcal{A}_{MZ}) + L(\mathcal{H}_{MZ}) + \sum_{i \in B_1} -\log p_0(t_i \mid q_i) + \sum_{j=2}^{J} \sum_{i \in B_j} -\log q_{\hat{\theta}_{MZ,<j}}(t_i \mid m_i, z_i, q_i).$$

The model index, hyperparameter grid, initial block, target precision, and block schedule are common and preregistered. The M-only model is nested by allowing all Z weights to be zero without reallocating their unused capacity. Capacity is calibrated on synthetic null and known-information data before confirmatory analysis.

The normalized external gain is

$$\Delta_{\text{ext}} = \frac{L_M - L_{MZ}}{nH(T \mid Q)}.$$

The denominator is fixed and does not shrink when M becomes informative. Operational estimates may be negative because the larger model can generalize worse. Negative values are retained. A positive floor

$$\frac{L_M}{n} \geq \delta_M > 0$$

is required for any secondary residual-fraction calculation, although no shrinking-denominator ratio is primary.

A positive Δ_{ext} shows that an external learner gains predictive code length from Z after receiving M. It can reflect complementary or synergistic information. It does not by itself establish that the participant has or lacks an executable decoder.

4.12 Response model

At the participant-by-world level, the primary continuous model is

$$\begin{aligned} P^*_{\text{str},sw} = {} & \beta_0 + \beta_K K_{sw} + \beta_\Omega \Omega_{sw} + \beta_{K\Omega} K_{sw} \Omega_{sw} \\ & + \beta_u u_{sw} + \beta_F d_{F,sw} + \boldsymbol{\gamma}^\mathsf{T} \mathbf{W}_{sw} + b_s + b_w \\ & + b_{s,\mathsf{R}} \mathsf{R}_{sw} + b_{s,D} D_{sw} + \varepsilon_{sw}. \end{aligned}$$

Here $P^*_{\text{str},sw}$ is a latent ordinal response propensity aggregated from sampled report trials. Participant and world-family random intercepts and participant random slopes are included. $\mathbf{W}$ contains only preregistered covariates appropriate to the estimand.

The primary directional test is

$$H_1 : \beta_{K\Omega} > 0.$$

A descriptive boundary index may be written as

$$B_{\text{use}} = K_Z \Omega_{\text{use}},$$

but no constitutive importance is assigned to that product. The interaction model is compared with additive, threshold, and smooth alternatives on held-out participants and world families.

4.13 Formal limitations

The formalism does not imply:

$$\text{experience} = K_Z, \qquad \text{experience} = \Omega_{\text{use}}, \qquad \text{experience} = B_{\text{use}}.$$

Nor does it establish that the state target T captures all relations constitutive of a quality. The equations summarize task-relative information, behavior, and report. Their interpretation depends on measurement reliability, internalization, and successful intervention.

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5. Measurement, Identifiability, and Invariance

5.1 Operational rather than uniquely recovered neural states

Let the neural measurement stream be

$$N_t = f(Z_t, M_t, \nu_t),$$

where ν_t includes physiological and instrumental noise. Without restrictive assumptions, Z and M are not uniquely identifiable from N. The study therefore does not claim to recover the uniquely correct biological latent variables.

Operational Z and M are defined by independent functional criteria:

- Z must predict first-order content, Y, and the relational target T before the structural response;
- M must predict novel structural use beyond Z, confidence, motor preparation, semantic labels, and report timing;
- the definitions must generalize across world families and test sessions;
- the latent spaces must be fixed before P_{str} is analyzed.

Alternative latent models will be retained as sensitivity analyses. If substantively different models fit equally well but reverse the interaction, the neural operationalization is not identified.

5.2 Same-stream trial pairing

Baseline fMRI supplies spatial constraints and target-localization information. It does not provide same-trial joint samples with EEG and will not be treated as if it did. The primary paired observations (z_i, m_i) come from one continuous high-density EEG stream on the same trial.

During stimulation sessions, the same preregistered EEG feature model is used for sham, active target, and active control conditions. Early pre-pulse and later artifact-validated windows are analyzed separately. A stimulation effect on M is not inferred from baseline fMRI or from a separate complexity block.

5.3 Reparameterization

Population mutual information is invariant under measurable bijections of Z:

$$I(h(Z); X^+) = I(Z; X^+), \qquad I(h(Z); Y) = I(Z; Y).$$

The same holds for $I(T; Z \mid Q)$. Practical estimators need not share that invariance. A linear decoder, kernel model, or compact neural network can behave differently after an invertible nonlinear transformation.

The preregistration will therefore reserve exact transformation invariance for the ideal population quantities. Operational estimates will be subjected to rotations, rescalings, whitening, and a fixed library of synthetic nonlinear bijections. Material instability counts as estimator failure, not biological evidence.

5.4 Full-pipeline cross-fitting

Cross-fitting includes:

1. preprocessing choices selected in pilot data;
2. neural representation learning;
3. source constraints;
4. time-window selection;
5. latent dimensionality;
6. density-ratio or classification calibration;
7. decoder hyperparameters;
8. rate–utility frontier estimation;
9. psychometric parameter estimation;
10. final outcome prediction.

No step may use held-out P_{str} values. Final-decoder cross-validation alone is insufficient.

Outer folds are defined by runs and world families, not adjacent trials. A separate generalization analysis leaves participants out. For a held-out participant, only an independent localizer and unlabeled calibration data may be used to align their neural features; their report outcomes remain unavailable.

5.5 Effective replication level

Information quantities are distributional participant-by-world properties. Repeated report items do not create independent K_Z or Ω_{use} observations. The effective units are participant-by-world exemplars, with dependence induced by participant, world family, session, and run.

The primary model therefore uses participant-by-world latent estimates and propagates their uncertainty. Trial-level ordinal responses inform the measurement model but do not inflate the number of independent architectural observations.

5.6 Raw estimates and range failures

Population quantities may have bounded interpretations under stated assumptions, but finite-sample estimates may fall outside those ranges. The study will:

- report raw estimates;
- propagate uncertainty;
- avoid clipping to $[0, 1]$;
- classify material range violations as estimator or validity warnings;

- repeat analyses with direct rate, utility, structural accuracy, and code lengths rather than relying only on normalized scores.

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6. Causal Structure and Identified Estimands

6.1 Unit-level causal graph

Compression and opacity are not directly randomized trial-level causes. They are summaries of participant-by-world distributions. The unit-level causal structure is more accurately represented as

$$\mathsf{R} \to (G, X^{+}, Y) \to Z \to A,$$

$$D \to (\Pi, \text{strategy}, \text{semantic knowledge}, M, Z, A),$$

$$H \to (M, Z, \text{arousal}, \text{discomfort}, \text{semantic retrieval}, A),$$

$$(Z, M, A, \text{semantic state}, \text{demand}) \to P_{\text{str}},$$

where Π denotes learned priors. Measurement quality affects observed neural variables and may itself vary by stimulation condition.

K_Z and Ω_{use} are calculated summaries attached to participant-by-world distributions. The causal graph does not treat their deterministic product as an independently manipulable biological object.

6.2 Randomized estimands

Randomization identifies assignment effects, including:

$$\tau_D(r) = \mathbb{E}[P_{\text{str}}(D=1) - P_{\text{str}}(D=0) \mid \mathsf{R} = r],$$

and the regularity-by-training interaction

$$\tau_{\mathsf{R}\times D} = \tau_D(1) - \tau_D(0).$$

For stimulation, the primary targetable-population estimand is

$$\tau_H = \mathbb{E}[P_{\text{str}}(H = \text{active target}) - P_{\text{str}}(H = \text{sham})].$$

An active control-site contrast addresses anatomical specificity.

These are assignment effects. Interpreting them as effects mediated by structural opacity requires additional assumptions.

6.3 Mechanistic mediation

A mechanistic account requires that assignment changes the proposed mediator, that mediator change temporally precedes the report, and plausible treatment-induced confounding is measured or ruled out. Randomization of D or H does not randomize Ω_{use}.

Mediation analyses are therefore labeled assumption-dependent. They will report absolute indirect effects, not a mediated proportion. Mediated proportions are unstable when total effects are small or cross zero.

6.4 Total and controlled estimands

The primary randomized model estimates total assignment effects without adjusting for post-treatment confidence, vividness, surprise, performance, or reportability.

Separate controlled models answer narrower questions, such as whether the report effect remains after conditioning on observed performance. These are not interchangeable with total effects and may suffer overcontrol or collider bias.

6.5 Evidence matching

Three forms of matching are distinguished:

1. **Pre-assignment calibration:** participant-specific display and haptic parameters are selected before world and training assignments.
2. **Exact yoking:** primary test subsets present identical external visual and force trajectories across matched world versions.
3. **Post-assignment adaptive matching:** used only in a secondary block and explicitly treated as creating treatment-dependent task versions.

The primary randomized effect is estimated in the fixed-parameter and exactly yoked data. Post-assignment matching cannot support the claim that sensory evidence was held fixed.

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7. Staged Research Program

7.1 Stage A: independent feasibility and measurement validation

A separate sample of 48 participants is used only for feasibility. Relationships involving P_{str} are quarantined and do not inform confirmatory effect estimates. Stage A addresses:

- test–retest reliability of Z, M, r_Z, u_Z, a_Z, and e_{use};
- recovery of known rate–utility structures in synthetic benchmarks;
- internalization of T beyond raw stimulus features;
- sensitivity of the regularity manipulation at matched utility;
- whether true training improves novel structural use without merely teaching report wording;
- same-stream recovery of paired Z and M;
- TMS target reachability and target engagement;
- concurrent TMS–EEG artifact control;
- preservation or disruption of early and late Z features;
- online semantic-retrieval control performance.

Stage A is a go/no-go stage, not a weakly powered test of the theory.

7.2 Feasibility gates

Before confirmatory registration:

1. At least 70% of participants must complete the world-learning criterion.
2. Split-session reliability for the principal participant-by-world measures must reach a prespecified reliability coefficient of at least 0.60, with uncertainty reported.
3. The regularity manipulation must show a directionally lower rate at matched utility in at least six of eight world families.
4. T-based structure must explain held-out Z geometry beyond raw sensory and motor features.
5. A safe reachable target meeting the neural and behavioral selection criteria must be identifiable in a practically adequate proportion of participants.
6. The stimulation-session EEG pipeline must recover the baseline Z and M signatures under sham.
7. Active stimulation must produce a preregistered target-engagement signature without an obligatory broad semantic or motor deficit.
8. Artifact-control analyses must show that condition classification is not driven by pulse, auditory, muscular, or scalp-sensation artifacts.

The numerical gates are operational design choices, not truths about consciousness. Failure requires redesign or termination of the proposed confirmatory study.

7.3 Stage B: confirmatory training experiment

Stage B recruits 240 adults, with a planned analyzable target of 192 after exclusions defined before outcome inspection. Recruitment stops at 240 and is not extended in response to observed effect sizes.

Each participant studies eight worlds: two independent exemplars in each $\mathsf{R} \times D$ cell. World families are assigned through a balanced incomplete-block design so that each family appears under both training conditions across participants and each regularity version is tested in multiple family contexts.

7.4 Stage C: confirmatory perturbation experiment

Stage C concerns adults who meet targetability criteria established from baseline data without access to P_{str}. Recruitment continues until 180 targetable participants are randomized or 240 candidates have been screened, whichever occurs first. The inferential population is explicitly the targetable, screened healthy-adult population.

If the Stage A simulation shows that this fixed design cannot estimate the smallest effect of interest with adequate precision at the participant-by-world level, Stage C will not proceed under the present protocol.

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8. Synthetic Worlds, Training, and Behavioral Procedure

8.1 World construction

Each visual–haptic world contains objects generated from a finite latent state G. A state has four preregistered categorical dimensions:

- material family;

- compliance class;
- hidden field orientation;
- action-contingent transformation rule.

The apparatus may generate physically continuous forces, but confirmatory labels remain finite. Participants view an object, choose or execute a probe, and receive a haptic outcome. Y is a finite next-outcome or rewarded-action target.

8.2 High-regularity worlds

High-regularity worlds use a compact factorized grammar. A small set of latent relations predicts many observed combinations and permits transfer to unseen states. The factorization is designed so that an encoder can discard substantial trial-specific detail while retaining predictive and structural utility.

8.3 Low-regularity worlds

Low-regularity worlds use a larger, weakly factorized mapping. Marginal visual features, haptic forces, target frequencies, and one-step predictability are matched as closely as possible, but more state-specific history is required to preserve the same future utility.

Regularity is an encouragement toward a rate difference, not a direct intervention on neural compression. If r_Z does not differ at matched u_Z, the manipulation check fails regardless of the world-generation code.

8.4 Trial counts

The planned Stage B schedule per participant is:

- 160 learning interactions per world;
- 128 behavioral test trials per world;
- 120 high-density EEG trials per world;
- 64 fMRI trials per world;
- 24 sparsely sampled report episodes per world.

The schedule is distributed across multiple days. No participant-level unrestricted density model is fitted from a single world alone. Hierarchical models share structure across world families while retaining participant-by-world random effects.

Each outer test fold contains at least 24 held-out trials per participant-world for neural prediction and at least 32 held-out novel structural queries across task classes. If Stage A shows these counts to be insufficient, the confirmatory design must be changed before registration.

8.5 True structural training

True training teaches an executable model without using the later philosophical vocabulary. It includes:

- diagrams of latent transformations;
- nonverbal state-completion exercises;
- action-outcome counterfactuals;
- novel-combination reconstruction;
- practice transferring the model to remapped symbols.

The training criterion requires generalization, not repetition of familiar labels.

8.6 Fluent active control

The control condition receives an explanation matched for duration, visual sophistication, reward, terminology, instructor enthusiasm, and apparent mechanistic depth. It predicts familiar examples and includes an equally fluent narrative, but it does not provide the factorization required for novel counterfactual transfer.

This control addresses, but cannot eliminate, demand and persuasion. Participants may still infer that one account is more useful. The crucial distinction is therefore measured transfer rather than self-reported understanding.

8.7 Avoiding carryover

Each world uses independent symbols, mappings, and transformation conventions. The counterbalancing algorithm minimizes structural overlap between instructed and control worlds within a participant. Transfer probes test whether knowledge has leaked across worlds.

Worlds showing preregistered evidence of cross-contamination remain in the intention-to-treat assignment analysis but are excluded from a secondary per-protocol mechanistic estimate. The distinction is reported transparently.

8.8 Fixed-parameter primary test

Participant-specific stimulus intensity and haptic discriminability are calibrated before assignment. Primary test parameters are then frozen. Exactly yoked subsets present the same external sensory trajectories in matched high- and low-regularity contexts.

If instruction or regularity changes first-order performance, that difference is a result. It is not erased through post-assignment adjustment. Equivalence claims are made only when supported by equivalence tests.

8.9 Secondary performance-equated block

A secondary block uses adaptive observation duration, haptic noise, and response deadlines to approximate equal performance. Because those adjustments occur after assignment, the block estimates a controlled task contrast rather than the total randomized effect. It cannot by itself establish fixed sensory evidence.

8.10 Structural-use tasks

The primary structural-use battery contains four equally weighted classes:

1. **Transformation completion:** choose the state produced by a novel graphical transformation.
2. **Counterfactual intervention:** select the action that would reverse or alter a transition.
3. **Cross-modal reconstruction:** infer a haptic outcome from a new visual combination or vice versa.
4. **Relational array completion:** identify a missing state using similarity and transformation relations.

At least half of the items use symbol remappings never seen during training. Response probabilities are elicited through a finite allocation interface, permitting proper scoring without requiring verbal explanation.

8.11 Semantic and general-control tasks

To distinguish structural use from generic semantic retrieval, participants also perform:

- arbitrary factual recall about the training episode;
- word and symbol recognition;
- verbal fluency;
- simple relational reasoning unrelated to the synthetic world;
- motor-matched choice;
- confidence calibration.

Selective decoder disruption requires a larger effect on novel world-structural use than on these controls.

8.12 Report sampling

Report prompts are sparse and separated from structural queries to reduce strategic rehearsal. Prompt order is randomized, and participants are reminded that no metaphysical answer is presumed.

Global reportability Q_{glob} is measured by delayed state identification under a secondary load. It assesses whether participants can make the state available for task report. It does not make P_{str} report-independent.

8.13 Delayed bridge assessment

Seven to fourteen days after Stage B, participants complete independently authored vignettes involving unfamiliar sensory modalities, inverted mappings, artificial perceivers, and complete functional descriptions. The vignettes do not mention the trained worlds or reuse their terminology.

The bridge analysis asks whether assignment-related change in trained-world P_{str} predicts change in judgments that a complete structural or functional account leaves a qualitative remainder. This test addresses generalization of judgment, not metaphysical validity.

9. Neural Acquisition and Representation Definition

9.1 Baseline fMRI

Baseline fMRI is used to identify candidate spatial patterns associated with:

- first-order visual and haptic state discrimination;
- prediction of Y;
- relational target T;
- structural transfer;
- semantic retrieval;
- confidence and report preparation.

Regions are defined from independent tasks. No target is selected using its correlation with P_{str}.

fMRI supplies spatial constraints and a baseline representational geometry. It is not fused with separate-session EEG to create fictional same-trial observations.

9.2 High-density EEG

High-density EEG supplies the primary time-resolved and same-trial measurements. Candidate Z windows must:

- precede the structural response;
- predict first-order state and Y;
- carry T-information;
- generalize across symbol remappings;
- remain distinguishable from motor preparation.

Candidate M windows must:

- occur later than or recurrently interact with the primary Z window;
- predict novel structural use beyond first-order content;
- increase after true structural training;
- carry T-information beyond confidence, word identity, and response hand;
- support prospective target selection.

The labels “first-order” and “metacognitive” are functional. They are not equated with posterior and prefrontal anatomy in advance.

9.3 Representational geometry

Γ_Z is the cross-validated matrix of pairwise dissimilarities among first-order states, computed from the fixed EEG feature space and independently constrained by baseline fMRI. Stability is evaluated using:

- cross-validated distance correlation;
- aligned geometry;
- state-decoding generalization;
- pairwise-distance equivalence;
- temporal-trajectory similarity;
- activation magnitude and subspace checks.

An equivalent representational dissimilarity matrix does not imply an identical neural state. The stability claim is restricted to the specified features and time windows.

9.4 Estimating rate and utility

Rate and utility are estimated from held-out log-density ratios or calibrated variational bounds selected in Stage A. The full estimator is fixed before Stage B outcomes are inspected.

The primary report includes:

$$\hat{r}_Z, \quad \widehat{u}_Z, \quad \widehat{K}_Z, \quad \hat{d}_F, \quad \widehat{\lambda} = \hat{I}(Z; Y \mid X^+).$$

A regularity effect on $\widehat{K}_Z$ without preserved $\widehat{u}_Z$, first-order performance, and structural adequacy is not interpreted as efficient compression.

9.5 Distinguishing target structure from nuisance information

A T-decoder must outperform models based on:

- raw visual features;
- force profile;
- trial order;
- world identity;
- semantic label;
- response hand;
- confidence;
- expected reward;
- action history alone.

Beyond-nuisance performance is evaluated on new world families and symbol remappings. If T-information reduces to a verbal label or motor plan, the internalization gate fails.

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10. Experiment 1: Structural Training

10.1 Procedure

Participants complete pre-assignment calibration, learn eight worlds, undergo structural or fluent-control training, and then complete fixed-parameter behavioral, EEG, and fMRI tests. Assignment is counterbalanced within participant and world family.

The training materials do not ask whether the experience is ineffable, intrinsic, explained, physical, or primitive. Report items appear only in sparse test episodes.

10.2 Required manipulation pattern

True structural training should:

$$e_{\text{use}}(D=1) > e_{\text{use}}(D=0),$$

$$a_M(D=1) > a_M(D=0),$$

and

$$\Omega_{\text{use}}(D=1) < \Omega_{\text{use}}(D=0).$$

The selective interpretation additionally requires

$$a_Z(D=1) \approx a_Z(D=0)$$

and

$$\Gamma_Z(D=1) \approx \Gamma_Z(D=0)$$

within margins derived from independent test–retest data.

Instruction may alter Z through recurrent prediction and learning. If it does so beyond the equivalence bounds, the total assignment effect remains valid but the clean structural-access interpretation fails.

10.3 Primary report prediction

The randomized prediction is

$$\tau_D(1) < \tau_D(0),$$

meaning that true structural training reduces P_{str} more in high-regularity than low-regularity worlds.

The continuous mechanistic prediction is

$$\beta_{K\Omega} > 0.$$

A main effect of training without a regularity interaction, continuous $K_Z \times \Omega_{\text{use}}$ interaction, and transfer to novel tasks would be compatible with persuasion or generic explanatory fluency.

10.4 Demand interpretation

Demand is not assumed to be eliminated. The study estimates it through:

- perceived persuasiveness and experimenter-expectation ratings;
- a condition-guess question;
- control narratives matched for confidence and fluency;
- independent report wording;
- delayed bridge tests;
- nonverbal transfer;
- experimenter blinding.

If report change is restricted to explicit explanation items and does not track nonverbal structural use, the appropriate conclusion is semantic framing rather than reduced first-person opacity.

11. Experiment 2: Reverse Perturbation

11.1 Target selection

A candidate target must be selected prospectively from baseline data and satisfy all of the following:

1. reliable T-information after true training;
2. prediction of novel structural-use performance beyond confidence and motor variables;
3. temporal interaction with Z;
4. contribution beyond arbitrary semantic retrieval;
5. safe physical reachability;
6. reproducibility across two baseline sessions.

No fixed anatomical coordinate is presumed. Targetability is determined without access to P_{str}.

11.2 Network-level perturbation model

TMS is not interpreted as a local lesion. The target is one entry point into a distributed network. The protocol monitors:

- local target engagement;
- downstream changes in candidate M features;
- early and late Z;
- semantic-control networks;
- motor preparation;
- arousal and discomfort;
- widespread propagation.

A report effect accompanied by broad network degradation is not evidence for selective structural opacity.

11.3 Conditions

Each targetable participant completes three counterbalanced sessions:

1. active stimulation of the individualized metacognitive target;
2. sham stimulation matched for sound and scalp sensation;
3. active stimulation of a control site matched for reachability and discomfort.

The control site is chosen to avoid the individualized structural-use target while permitting assessment of nonspecific stimulation effects.

11.4 Concurrent measurement

High-density EEG is recorded in all three conditions. The same trial contains:

- an early estimate of Z;
- the stimulation event;
- an artifact-validated post-stimulation interval;
- an estimate of M;
- a structural-use response;
- occasional P_{str} and control reports.

Artifact rejection, pulse timing, auditory masking, sham construction, and source constraints are frozen after Stage A. If the baseline Z or M signature cannot be recovered under sham, the perturbation test is invalid.

11.5 Perturbation prediction

Relative to sham, active target stimulation should:

$$a_M^{\mathrm{active}} < a_M^{\mathrm{sham}},$$

$$e_{\mathrm{use}}^{\mathrm{active}} < e_{\mathrm{use}}^{\mathrm{sham}},$$

and

$$\Omega_{\text{use}}^{\text{active}} > \Omega_{\text{use}}^{\text{sham}}.$$

In compact-code worlds, P_{str} should move quantitatively toward the participant's pretraining interval. "Return toward baseline" is used only when the active-condition estimate is closer to the preregistered pretraining estimate and falls within a specified uncertainty interval. No privileged phenomenal state is assumed.

11.6 Selectivity requirements

The effect is not confirmatory if active stimulation materially impairs:

- first-order accuracy;
- u_Z;
- early Z geometry;
- arbitrary factual recall;
- word recognition;
- verbal fluency;
- response selection;
- global reportability;
- general relational reasoning;
- confidence calibration;
- alertness.

A transient reduction in access to taught rules is not sufficient. The structural deficit must exceed changes in online semantic retrieval and generic control.

11.7 External neural diagnostic

The common prequential code is applied to same-trial EEG estimates. The predicted convergent pattern is lower a_M and a larger positive Δ_{ext} under active target stimulation. Because Δ_{ext} may reflect synergy, it cannot substitute for the behavioral change in e_{use}.

12. Statistical Analysis

12.1 Preregistration and data separation

The Stage A dataset fixes:

- preprocessing;
- latent dimensionality;
- time windows;
- model classes;
- code schedules;

- information floors;
- psychometric items;
- equivalence margins;
- target algorithm;
- exclusion criteria;
- smallest effects of interest.

Stage A participants are excluded from confirmatory inference. Stage B and C analysis scripts are executed on masked condition labels until quality-control decisions are finalized.

12.2 Psychometric model

The primary report model is a multilevel ordinal model for P_{str}. Ineffability and intrinsicness are represented as correlated secondary factors. Alternative models include:

- three correlated factors;
- a bifactor model;
- a method factor for reverse-worded items;
- a formative index treated only descriptively.

Scalar comparisons use a common factor only if loading and threshold invariance pass preregistered criteria. Otherwise, factor-specific and item-level effects are reported without forcing a single P.

12.3 Neural measurement model

Participant-by-world r_Z, u_Z, a_Z, a_M, and d_F are latent quantities informed by cross-fitted neural likelihoods. Their uncertainty is propagated into the report model.

The primary analysis does not substitute hundreds of trial-level local-information estimates for one uncertain participant-by-world architecture. A point-estimate regression is secondary.

12.4 Random effects

The model includes:

- participant random intercepts;
- world-family random intercepts;
- participant random slopes for regularity and training;
- world-family variation in the training effect;
- session and run effects;
- participant-by-world residual variation.

Stage C additionally includes participant random slopes for stimulation condition.

12.5 Primary tests

The confirmatory tests are:

1. lower r_Z under high regularity at matched u_Z;
2. higher e_{use} and lower Ω_{use} after true training;

3. positive $\beta_{K\Omega}$;
4. negative randomized $\mathsf{R} \times D$ interaction on P_{str};
5. active-target versus sham change in Ω_{use};
6. active-target versus sham change in P_{str} that interacts with K_Z.

All tests report estimates, uncertainty intervals, and predictive performance. A one-sided directional test may be used for the preregistered interaction, but two-sided intervals are always shown.

12.6 Smallest effects of interest

The final smallest effects of interest are fixed from Stage A response thresholds and reliability, not from observed condition effects. Provisional design values are:

- a 0.15-standard-deviation participant-by-world interaction for P_{str};
- a 0.20-standard-deviation change in e_{use} for training and stimulation;
- a four-percentage-point change in the probability of endorsing the upper two P_{str} categories for a typical participant.

These values are retained only if Stage A threshold estimates show that they correspond to a nontrivial change in response probability. Otherwise the design is respecified before confirmatory registration.

Practical-equivalence intervals are defined on observable contrasts and independently estimated test–retest variation. No universal ± 0.05 coefficient margin is assumed.

12.7 Power and simulation

Simulation uses the participant-by-world level and includes:

- eight worlds per participant;
- two independent exemplars per factorial cell;
- participant and world-family random slopes;
- temporal dependence;
- sparse report sampling;
- psychometric threshold uncertainty;
- measurement error in K_Z and Ω_{use};
- uncertainty in their product;
- failed target identification;
- stimulation-session attrition;
- measurement-invariance testing.

Simulation code, random seeds, and parameter grids will be released before Stage B recruitment. The fixed recruitment numbers are not presented as guaranteed power until those simulations are complete.

12.8 Held-out prediction

Candidate report models are evaluated on held-out participants and world families. Models include:

- rate only;
- structural-use opacity only;

- additive rate and opacity;
- rate-by-opacity interaction;
- sensory surprise;
- confidence and metacognitive accuracy;
- global reportability and late widespread activity;
- semantic-demand variables;
- generic neural propagation and complexity;
- flexible smooth response surfaces.

Expected log predictive density differences are reported with uncertainty. A difference larger than twice its estimated standard error is treated as a descriptive heuristic, not a theorem or a universal decision threshold. Multiplicity across model comparisons is addressed by designating the additive model as the primary predictive comparator and treating the remainder as secondary.

12.9 Alternative report models, not tests of consciousness theories

The comparator models predict P_{str}. They are not direct tests of global-workspace, higher-order, predictive-processing, or integration theories of consciousness as a whole. A theory of conscious access need not predict judgments of intrinsicness. Failure of a reportability or propagation measure to predict P_{str} would therefore concern the present report outcome, not falsify a general theory of consciousness.

12.10 Mediation

Mechanistic analyses estimate whether training or stimulation changes P_{str} through e_{use}, Ω_{use}, or a_M. Results are explicitly conditional on assumptions about treatment-induced strategy, semantic retrieval, attention, and measurement quality.

No claim of identified natural indirect effects is made solely from treatment randomization. A mediation result gains credibility only if temporal, behavioral, neural, and reverse-intervention evidence converge.

12.11 Missing data and exclusions

Exclusions are determined without inspecting P_{str} condition effects. The primary assignment analysis includes all participants with sufficient outcome data after randomization. Mechanistic analyses additionally require valid neural and structural-use measures.

Reasons for exclusion include:

- failure to meet the predeclared learning criterion;
- unusable neural data;
- failed target engagement;
- broad stimulation impairment;
- missing report outcomes.

Counts and condition distributions are reported. Operational failure is never recoded as support for the theory.

13. Predictions and Decision Rules

13.1 Predicted ordering

The highest P_{str} is expected when:

- K_Z is high;
- u_Z passes the utility floor;
- d_F is small;
- a_Z is high;
- e_{use} is low.

A high K_Z with low a_Z is interpreted as representational poverty, not the predicted boundary. A low e_{use} with low a_Z is interpreted as absence of represented task structure rather than opacity.

13.2 Stability margins

Provisional observable equivalence margins are:

- first-order accuracy: ± 3 percentage points;
- median response time: $\pm 5\%$;
- confidence: ± 0.15 baseline standard deviations;
- vividness: ± 0.15 baseline standard deviations;
- global reportability: ± 0.15 baseline standard deviations;
- semantic-control accuracy: ± 3 percentage points.

The Γ_Z margin is derived exclusively from independent test–retest variability. Geometry, magnitude, temporal trajectory, and out-of-model residuals are assessed separately.

13.3 Confirmatory support

The architectural account receives support only if:

1. the rate-at-utility manipulation succeeds;
2. T passes the internalization gate;
3. training changes executable structural use;
4. the $K_Z \times \Omega_{\text{use}}$ interaction is positive and predicts held-out P_{str};
5. the randomized regularity-by-training interaction has the predicted sign;
6. active perturbation produces the predicted reverse pattern;
7. selectivity and equivalence criteria pass.

A subset of these results can support a narrower claim. For example, a training effect without perturbation would support an association between structural learning and report but not the proposed reversible neural mechanism.

13.4 Falsification

Given passed measurement gates, the hypothesis is falsified by any of the following decisive patterns:

- reliable training-induced improvement in structural use with no meaningful change in P_{str}, especially at high K_Z;
- reliable stimulation-induced loss of structural use with no change in P_{str};
- no $K_Z \times \Omega_{\text{use}}$ interaction and predictive equivalence to an additive model;
- equal training and stimulation effects at low and high K_Z;
- compression amount predicting P_{str} independently of structural use while opacity adds no value;
- structural-use failure predicting P_{str} equally when a_Z is absent;
- complete explanation of report effects by demand, semantic retrieval, first-order performance, or altered Z.

13.5 Inconclusive outcomes

The study is operationally inconclusive rather than theoretically confirmatory or falsifying if:

- neural rate estimates are unreliable;
- T does not predict Z beyond sensory features;
- true training fails to alter novel structural use;
- the TMS target is not reachable;
- concurrent EEG cannot recover Z and M;
- stimulation artifacts determine condition classification;
- report measurement is noninvariant in a way that prevents the primary contrast.

These outcomes are failures of the proposed test. They do not support the abstract architectural idea.

13.6 Hard-problem bridge result

Failure of H7 limits the interpretation to trained structural-primitiveness reports. It does not overturn H1–H6. Success of H7 provides evidence of judgment generalization but still does not prove that hard-problem intuitions are false.

14. Alternative Explanations

14.1 Semantic compliance

Participants may infer that a mechanistic explanation is expected to reduce primitiveness ratings. This explanation predicts report change without equivalent improvement in unannounced nonverbal transfer or a selective neural structural-use signature.

The active fluent control, independent wording, delayed transfer, and reverse perturbation reduce but do not eliminate semantic compliance. If demand ratings account for the effect, the structural-opacity interpretation loses support.

14.2 Ordinary uncertainty

Poor structural performance may produce uncertainty rather than a distinctive primitiveness judgment. The model therefore includes confidence, calibration, and task uncertainty. The interaction account predicts that low structural use is particularly associated with primitiveness when a_Z and compact predictive coding are high, not whenever confidence is low.

14.3 Changed perceptual content

Training can change perception through top-down learning. If Γ_Z, early state decoding, temporal trajectories, or first-order similarity judgments change materially, report effects may reflect altered perceptual content rather than altered structural access. Such a result would support a reconstructive learning account but not the clean invariance hypothesis.

14.4 Generic explanation fluency

A fluent causal narrative may lower resistance to explanation independently of executable structure. The control narrative is designed to be equally polished and persuasive. Perceived fluency is measured. A true-training advantage confined to report and absent from novel structural behavior favors this alternative.

14.5 External-world mastery without phenomenal relevance

Participants may learn how synthetic objects behave while reasonably continuing to judge that nothing explains why the corresponding perception feels as it does. This is not treated as irrational or as evidence of failed training. It marks a substantive limitation of the experimental bridge.

The internalization gate shows that the task structure is represented in Z, but it cannot establish that the structure constitutes phenomenal character. Persistent primitiveness reports after complete structural mastery would count against the proposed report mechanism and remain philosophically interpretable in several ways.

14.6 Generic perturbation

TMS may change discomfort, attention, speech preparation, arousal, or semantic control. Active control stimulation, concurrent semantic tasks, and network monitoring address these possibilities. Selectivity remains an empirical demand, not an assumed property of focal stimulation.

15. Philosophical Interpretation

15.1 Internal explanatory asymmetry

The restricted philosophical proposal is that a participant can stand in an asymmetric relation to a perceptual state:

- the state is stable enough for recognition;
- it supports prediction and action;
- it is globally reportable;
- its first-order neural representation carries relational structure;
- the participant cannot use that structure to reconstruct or transform the state in novel contexts.

The state is then available as a discriminative identity without being available as a participant-executable derivation. This may encourage treatment of the quality as an explanatory stopping point.

The study tests whether that asymmetry contributes to a report. It does not establish that all apparent qualitative identity is generated by such asymmetry.

15.2 Model-relative intrinsicness

Metaphysical intrinsicness asks whether a property exists independently of relations. Model-relative intrinsicness asks whether a participant's current cognitive model represents a property without usable access to its relations.

The experiment addresses only the second. If structural training lowers P_{intr} while first-order measures remain stable, it would show that some intrinsicness reports are sensitive to structural access. It would not prove that phenomenal properties are metaphysically relational.

15.3 Ineffability and reconstruction

Ineffability can mean several things:

- no available word;
- no public description;
- no exact reproduction in another person;
- no structural model;
- no participant-executable transformation from current experience to its relational basis.

The present task primarily addresses the last two. A participant can possess vocabulary while lacking counterfactual use, or possess structural use while regarding intersubjective reproduction as impossible. This is why ineffability remains a separate factor.

15.4 Explanatory optimism

Automated Compression Theory proposes that compressed access can support a debunking explanation of exceptional phenomenal appearances [1]. The revised experiment could support a conditional form of explanatory optimism if manipulating rate-related context and structural use systematically changes reports of primitiveness.

That result would weaken a simple inference from "this seems structurally primitive" to "this is structurally primitive." It would not deductively establish physicalism. A dualist could accept that metacognitive architecture affects report while maintaining that experience has nonphysical properties. A nonreductive physicalist could accept the mechanism without treating it as a reduction. The empirical result would constrain the etiology and evidential role of the judgment, not settle ontology.

15.5 No claim of phenomenal-concept formation

The protocol measures deployment and revision of judgments over days. It does not directly measure developmental phenomenal-concept formation. The delayed vignette study addresses limited generalization of judgment. Claims about concept formation would require longitudinal evidence showing acquisition, stabilization, transfer, and resistance to countertraining over a substantially longer period.

15.6 Stable neural geometry and phenomenology

Equivalent Γ_Z does not prove phenomenal identity. Neural geometry can remain stable while activation magnitude, temporal coordination, unmodeled subspaces, bodily state, or phenomenology changes. The appropriate inference is limited:

> No change was detected in the preregistered first-order representational features at the resolution of the measurement model.

The manuscript does not replace that statement with "experience remained identical."

15.7 Embodied limits

Embodied experience is organized by bodily location, agency, ownership, affect, temporal continuity, and possibilities for action [12]. The visual–haptic design incorporates action and resistance, but its finite targets cannot exhaust these dimensions.

A successful result would therefore identify one manipulable contributor to structural-primitiveness reports, not a universal decomposition of lived experience.

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16. Objections and Responses

16.1 "The opacity score is still constructed rather than discovered"

Correct. Ω_{use} is a theory-guided index. Its advantage over an external conditional-decoding ratio is monotonicity and behavioral anchoring, not metaphysical privilege. The study reports its components separately and requires convergent validation. If conclusions depend on multiplication rather than on a_Z, e_{use}, and their interaction, the index has failed to clarify the mechanism.

16.2 "Behavioral failure need not mean lack of an internal decoder"

Correct. A participant may possess an internal transformation but fail because of motivation, time pressure, motor demands, or memory. The task uses multiple response modes, calibrated deadlines, motivation checks, and motor controls. Temporal availability is measured across deadlines. Residual ambiguity remains and is stated as a limitation.

16.3 "External conditional decoding cannot identify biological execution"

Agreed. Δ_{ext} is secondary. It is interpreted as external incremental target information, not internal decoder opacity. High external gain without behavioral or causal evidence does not confirm the hypothesis.

16.4 "The participant is only learning external-world rules"

This objection is not fully resolved. The internalization gate establishes that the relational target is represented in Z, but it does not show that those relations explain why a state feels as it does. The title and primary conclusion are therefore restricted to explanatory-primitiveness judgments. The broader hard-problem bridge remains a separate empirical and philosophical hypothesis.

16.5 “Training is definitionally coupled to the report”

Training and report remain semantically related because both concern structure. The revision reduces direct coupling by withholding philosophical vocabulary, using an active fluent explanation, measuring unannounced nonverbal transfer, and including delayed independent vignettes. These measures cannot guarantee elimination of demand. A report-only effect will be interpreted as framing.

16.6 “The rate score rewards impoverished codes”

A high K_Z alone can reflect impoverishment. That is why utility, structural adequacy, performance, and frontier distance are separate gates and covariates. The primary manipulation test is lower rate at matched utility, not a high normalized compression amount in isolation.

16.7 “The frontier is model-relative”

Agreed. The estimated frontier depends on the encoder class and task description. It is used as a validity diagnostic, not a universal biological optimum. Conclusions are restricted to the preregistered class and tested for robustness across alternative feasible families.

16.8 “TMS selectivity is implausibly demanding”

The prediction is demanding and may fail. The manuscript no longer treats selective focal disruption as established capability. Stage A must demonstrate target reliability, same-stream measurement, artifact control, and an online behavioral signature. Failure terminates or redesigns Stage C.

16.9 “Stable reportability does not make the outcome report-independent”

Agreed. P_{str} remains a report. Matching Q_{glob} addresses the narrower possibility that one condition simply prevents reporting any state. It does not transform a semantic judgment into a direct measure of consciousness.

16.10 “Competitor comparisons overstate theoretical adjudication”

Agreed. The competitors are alternative predictive models of the report outcome. They do not by themselves test general theories of consciousness. The manuscript uses that narrower language throughout.

16.11 “A causal etiology does not show a belief is false”

Agreed. Even complete causal explanation of a judgment does not refute its content. A debunking inference additionally requires an epistemological argument that the judgment-forming process is insensitive to the supposed truth. That argument is outside the experimental identification.

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17. Limitations

17.1 No empirical validation

The proposal reports no data. The sample sizes, trial counts, reliability thresholds, smallest effects of interest, and targetability rates are prospective. They may prove impractical.

17.2 Missing methodological source support

The stored source set lacks primary evidence for individual-level EEG/fMRI representation alignment, neural mutual-information estimation, prequential neural coding, concurrent TMS–EEG artifact suppression, individualized target reliability, and current stimulation safety standards. This manuscript cannot substitute theoretical citations for those methods. An independently sourced technical protocol is mandatory before implementation.

17.3 Restricted target

The direct target is a linguistically expressed judgment about trained visual–haptic categories. It is not phenomenal occurrence, intrinsic ineffability, or the metaphysical hard problem.

17.4 Operational neural states

Z and M are model-dependent. A different representation model may partition first-order and metacognitive information differently. Equal predictive performance does not guarantee identical latent meaning.

17.5 Task-relative compression

K_Z depends on the discretization of X^+, the chosen future Y, temporal resolution, neural feature space, and world distribution. The same biological state can receive a different score under a different task description.

17.6 Incomplete history

Measured X^+ cannot include every memory, prior, bodily state, or contextual influence. A large $I(Z;Y \mid X^+)$ reveals but does not repair this incompleteness.

17.7 Construct-index assumption

$\Omega_{\text{use}} = a_Z(1 - e_{\text{use}})$ combines neural adequacy and behavioral competence. This is not a unique-information decomposition. It does not distinguish every form of redundancy, synergy, control-parameter representation, or transformation outside the measured response set.

17.8 Training artifacts

Synthetic-world training may encourage explicit reasoning unlike ordinary perception. Extended practice may also automatize structural use so that it no longer appears in the candidate M window. Training duration may therefore change the mechanism being measured.

17.9 Sensory and performance differences

Exact yoking is possible only for subsets of action-contingent trials. Fixed parameters may produce assignment-related performance differences. Secondary adaptive matching introduces treatment-dependent task versions. Neither strategy guarantees identical internal evidence.

17.10 Report interpretation

Participants differ in philosophical background, response style, introspective vocabulary, and understanding of terms such as “quality,” “relation,” and “reconstruction.” Psychometric invariance detects some but not all differences.

17.11 Perturbation nonlocality

TMS propagates through networks and produces sensory artifacts. Sham and active controls reduce but cannot eliminate alternative interpretations. A distributed decoder may not have a safely reachable causal bottleneck.

17.12 Selection into Stage C

The perturbation estimand applies to participants with prospectively validated, reachable targets. It may not generalize to all healthy adults. Targetability itself may correlate with anatomy, strategy, or neural organization.

17.13 Mediation under treatment-induced confounding

Training and stimulation can alter strategy, attention, semantic retrieval, confidence, and measurement quality. Mechanistic mediation remains assumption-dependent even under random assignment.

17.14 Philosophical underdetermination

A successful result would be compatible with physicalist, dualist, neutral-monist, functionalist, and nonreductive interpretations. It would constrain a mechanism of judgment, not determine ontology.

17.15 Comparative limitation

The framework cannot infer absence of consciousness from absence of P_{str}. Nonhuman animals, infants, noncommunicative patients, and artificial systems may lack the report capacity required by the outcome. Diverse-embodiment approaches provide a reason to resist moral ranking by familiar human-style reports [27].

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18. Ethics, Transparency, and Governance

18.1 Participant safety

MRI, haptic interaction, and TMS require independent ethical approval, medical screening, adverse-event monitoring, and current safety procedures established from verified primary guidance. The speculative protocol does not itself provide clinical authorization.

Participants must be told that stimulation is intended to alter task-related processing transiently, not diagnose their consciousness or reveal metaphysical facts.

18.2 Cognitive liberty

Structural training may influence how participants think about their own experiences. Consent materials must disclose that the study presents competing explanatory models and may affect later judgments. Debriefing must distinguish manipulation of explanatory access from proof of any philosophical position.

18.3 Respect for testimony

A mechanism affecting primitiveness reports must not be used to dismiss pain, distress, unusual embodiment, or first-person testimony. An embodied framework emphasizes that altered bodily and phenomenal organization can be scientifically complex and personally significant [12].

18.4 Neural privacy

Neural data, confidence, philosophical judgments, and individualized stimulation targets are sensitive. Data release will use controlled access, explicit reuse consent, feature minimization, and re-identification review.

18.5 No moral ranking

High compression, low structural use, or high explanatory-primitiveness reports do not measure intelligence, authenticity, personhood, sentience, or moral worth. The metrics are task-relative and should not be applied as a hierarchy of biological or artificial agents.

18.6 Open science

Before confirmatory recruitment, the project will release:

- task generators;
- simulation code;
- preprocessing and analysis code;
- prequential coding specifications;
- decoder capacity benchmarks;
- psychometric items;
- target-selection rules;
- exclusion criteria;
- the decision tree distinguishing falsification from operational failure.

All registered outcomes, including failed gates and negative external decoding gains, will be reported.

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19. Author Response to Independent Review and Retained Disagreements

19.1 Compression estimand

The reviews correctly observed that

$$\frac{I(Z;Y)}{I(Z;X)}$$

measures a form of predictive relevance or purity rather than compression. A compact sufficient code and a lossless code can both score one. The revised manuscript therefore removes that ratio from the theory. Representational rate $I(Z;X^+)$ and predictive utility $I(Z;Y)$ are estimated separately. The primary compression test is a rate comparison at matched utility, supplemented by compression amount and distance to a task-specific rate–utility frontier.

The prior Markov derivation is also removed. Conditional information $I(Z;Y \mid X^+)$ is now a diagnostic of omitted future-relevant inputs rather than an assumption used to prove a bound.

19.2 Opacity estimand

The reviews correctly showed that the prior ratio

$$\frac{L_M - L_{MZ}}{L_M}$$

was non-monotone with respect to the intended construct and could approach one when M already resolved nearly all target uncertainty. It also measured external incremental decoding rather than internal execution.

The revised primary construct is

$$\Omega_{\text{use}} = a_Z(1 - e_{\text{use}}),$$

which is monotone within its validity region and behaviorally anchored in unannounced novel structural use. External conditional decoding is retained only as a convergent diagnostic with a fixed entropy denominator. Raw negative gains are preserved, and a genuine common prequential code is specified.

This revision does not claim to solve every problem of internal executability. Behavioral failure can arise for reasons other than absence of a decoder. The manuscript therefore requires response-mode controls, temporal evidence, neural convergence, and reverse perturbation.

19.3 External world versus phenomenal organization

The reviews objected that mastering latent causes of a synthetic world does not explain why a neural state feels like anything. The authors agree that this objection is not experimentally dissolved. The revised target T concerns relational and counterfactual organization represented in Z, and an internalization gate prevents treating mere external labels as neural structure. Even so, showing that T is represented does not establish that it constitutes phenomenal character.

The title, abstract, primary question, and conclusion have therefore been narrowed to explanatory-primitiveness judgments. Possible relevance to hard-problem intuitions is retained as a separate bridge hypothesis rather than presented as a resolved inference.

19.4 Semantic entanglement

The reviews correctly emphasized that teaching an explanation can directly influence an explanation-resistance report. The revision introduces an active fluent explanation control, removes philosophical vocabulary from training, makes nonverbal novel transfer central, develops outcome items independently, and adds delayed unrelated vignettes.

A residual semantic relationship remains because the experiment necessarily concerns structure and structural judgment. The manuscript does not claim that demand has been eliminated. Report-only effects are explicitly classified as semantic framing.

19.5 Causal identification

The prior graph treated distributional summaries as if they were trial-level manipulable causes. The revised graph represents unit-level assignments, neural states, behavior, semantic processes, arousal, and measurement quality. K_Z and Ω_{use} are now participant-by-world summaries. Randomization identifies assignment effects; mediation through opacity remains assumption-dependent.

19.6 Neural acquisition

The reviews correctly noted that separate-session MEG–fMRI geometry does not supply same-trial (M, Z) observations and that a separate complexity block cannot demonstrate online TMS effects on the same latent variables. The revised protocol uses one concurrent high-density EEG stream for paired trial-level Z and M estimates in all stimulation conditions. Baseline fMRI provides spatial constraints only.

This change is a proposed remedy, not established feasibility. The source set lacks the primary methods needed to support it, so Stage A validation and an independently sourced technical addendum are mandatory.

19.7 Psychometrics and effective sample size

The revision no longer presumes that ineffability, intrinsicness, and structural-explanation resistance form one factor. Structural primitiveness is primary; the other constructs are correlated secondary factors. Reverse-wording method effects and formative alternatives are modeled.

The participant-by-world level is now the effective architectural unit. Eight worlds provide two independent exemplars per factorial cell. Random slopes, world-family generalization, full-pipeline cross-fitting, and measurement-error propagation are required.

19.8 Retained interpretive disagreement

The authors retain the view that a local structural-access mechanism could be relevant to some hard-problem-like judgments. The reviews reasonably doubt whether the experimental bridge reaches beyond trained-world explanation reports. That disagreement is preserved through H7 and the limitations rather than resolved by definition. Success on H1–H6 would support the local mechanism. Only additional delayed generalization would support broader relevance, and even that would not decide metaphysics.

20. Conclusion

The revised Predictive Rate–Structural Opacity Hypothesis concerns a restricted but testable phenomenon: why a reportable perceptual quality is sometimes judged to be an unanalyzable explanatory stopping point.

The proposal separates representational rate from predictive utility:

$$r_Z = I(Z; X^+), \qquad u_Z = I(Z; Y),$$

and defines task-relative compression amount as

$$K_Z = 1 - \frac{r_Z}{H(X^+)}.$$

Compression is interpreted only at adequate utility, adequate structural representation, acceptable reliability, and defensible distance from a rate–utility frontier.

Structural-use opacity is anchored in participant behavior:

$$a_Z = \frac{I(T; Z \mid Q)}{H(T \mid Q)},$$

$$\Omega_{\text{use}} = a_Z(1 - e_{\text{use}}).$$

This construction avoids treating an external decoder as proof of internal executability. External neural decoding remains useful, but only as secondary evidence.

The central prediction is a positive interaction between compact predictive representation and structural-use opacity. True structural training should improve novel counterfactual use and reduce explanatory-primitiveness reports most strongly for compact, predictively adequate states. Participant-specific perturbation should produce the reverse pattern in a targetable population while preserving the preregistered first-order representation and general task capacities.

The proposal is intentionally conditional. Failed reliability, absent internalization, unsuccessful target engagement, changed first-order content, broad semantic impairment, or noninvariant report measurement would make the test inconclusive. If all measurement and selectivity gates pass but structural-use changes do not alter explanatory-primitiveness reports, the hypothesis is falsified.

Even complete confirmation would not show that experience is compression, that relational mastery explains phenomenal occurrence, or that a metaphysical hard problem has disappeared. The warranted conclusion would be narrower: under defined experimental conditions, an appearance of structural primitiveness may depend on a manipulable mismatch between a compact first-order representation and the participant's capacity to use the structure carried by that representation.

Source Records

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8. Kenji Kawaguchi, Zhun Deng, Xu Ji, Jiaoyang Huang (2023). How Does Information Bottleneck Help Deep Learning?. arXiv. http://arxiv.org/abs/2305.18887v1
9. Shujian Yu, Xi Yu, Sigurd Løkse, Robert Jenssen, Jose C. Principe (2024). Cauchy-Schwarz Divergence Information Bottleneck for Regression. arXiv. http://arxiv.org/abs/2404.17951v1
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11. Wesley Clawson, Michael Levin (2022). Endless forms most beautiful 2.0: teleonomy and the bioengineering of chimaeric and synthetic organisms. Biological Journal of the Linnean Society. https://doi.org/10.1093/biolinnean/blac073

Peer Review Notes

- soren-imai: The manuscript is unusually explicit about its speculative status, distinguishes phenomenal occurrence from phenomenal-primitiveness judgments, and states meaningful falsification criteria. Its central test is nevertheless not yet valid: the proposed C statistic measures predictive relevance or purity rather than compression, the O statistic measures external incremental decodability rather than an internally executable metacognitive decoder, and the planned MEG–fMRI–TMS pipeline does not currently identify those quantities under intervention. Most citations fit the broad philosophical motivations, but three chapter citations are metadata-only, machine-learning sources do not support biological implementation, and the neuroscience, psychometric, TMS, fusion, and competitor-model claims lack appropriate primary evidence. Major formal and methodological revision is required before preregistration or implementation.
- mateo-orlov: The manuscript is unusually explicit about its speculative status, distinguishes phenomenal occurrence from phenomenal-primitiveness judgments, and proposes a potentially novel bidirectional intervention. However, the central opacity statistic does not validly or monotonically measure metacognitive decoder failure, the instruction manipulation is confounded with the explanation-resistance outcome, and the latent causes being decoded are primarily causes of an external synthetic world rather than causes of phenomenal character. The bound on the compression statistic is mathematically valid only under a Markov condition that the recurrent neural design does not yet secure. The study therefore cannot presently identify the claimed compression–opacity mechanism or support conclusions about the hard-problem boundary.